Group 4 EDA Video Presentation — Full Speech Script

🎤 MEMBER 1 — Opening, Dataset Overview & EDA Design

Target: 2:00 minutes (~280 words)

Hi everyone, I'm [Member 1 name], and along with [Member 2], [Member 3], and [Member 4], we're Group 4 from FIT5196 Data Wrangling. In the next nine minutes, we'll walk you through our Exploratory Data Analysis of a Flickr photo-metadata dataset, the ten findings we uncovered, and the five machine learning questions we've derived from them.

Let me start with the dataset itself. Our source data came in two files — an XML file and a JSON file — which together contained 32,892 photograph records, each with 18 attributes. These attributes range from simple identifiers like Post ID and User ID, to geospatial fields like latitude and longitude, to rich user-generated text like titles, tags, and descriptions.

In Step 1, we built a Python pipeline that merged both files, removed redundant duplicates, stripped XML and HTML tags, removed emojis and non-Latin characters using regular expressions, retained accented European characters, lowercased the five specified text attributes, and standardised all missing values to the string 'NaN'. The output was a single clean CSV that we used for every subsequent analysis.

We designed our EDA as a five-phase pipeline. Phase one was structural validation — checking dimensions, data types, and missingness. Phase two was univariate analysis across numeric, categorical, temporal, and textual attributes. Phase three was bivariate analysis — examining pairs of attributes for relationships. Phase four was multivariate analysis through compound visualisations. And phase five was insight synthesis — translating our findings into business and research value. The entire analysis was implemented in Python using Pandas, Matplotlib, Seaborn, and NumPy.

Now I'll hand over to [Member 2], who will walk you through our first set of findings on data quality and geography.

🎤 MEMBER 2 — Findings: Missing Data, Geography & Country Naming

Target: 2:15 minutes (~315 words)

Thanks [Member 1]. Hi everyone, I'm [Member 2], and I'll cover our findings on data completeness, geography, and country naming.

Our first finding was that the dataset is substantially incomplete — and the pattern is telling. Description is missing in 57.6% of records, City in 55.7%, and Country in 53.3%, while Tags is missing in 30.8%. But here's the interesting part: latitude and longitude are missing in only one record each. This tells us that users are consistently uploading GPS coordinates, but are skipping the textual location fields. This is a behavioural pattern, not a data collection failure.

Our second finding is that the dataset is geographically concentrated in Central and Western Europe. All geotagged records sit between 35 and 60 degrees latitude and 10 and 25 degrees longitude. Sweden leads with 2,195 records, followed by the USA, Italy, Poland, and Germany. The dataset clearly does not represent global Flickr usage.

But our third finding complicates the geography picture. Country names are entered as free text, and we found that Germany appears under two variants — "germany" and "deutschland" — totalling 1,879 records when combined. Italy appears as "italia" and "italy" — totalling 1,996 combined. This multilingual fragmentation artificially inflates the apparent diversity of the dataset and would distort any country-level modelling if left unnormalised.

Our sixth finding zooms in on the city level. Stockholm appears 1,717 times — which is 2.56 times more frequent than the second-ranked city, Washington DC. This dominance isn't fully explained by Sweden's country-level count, which suggests a concentrated user community centred in Stockholm during the dataset's collection window. It's a community concentration effect, not just a national one.

So the headline from this block: the data is geographically skewed, textually under-filled, and carries entity-level inconsistencies that any downstream model must address before training.

I'll hand over to [Member 3], who'll take you through user behaviour, temporal patterns, and tag content.

🎤 MEMBER 3 — Findings: Users, Time, Behaviour & Tags

Target: 2:15 minutes (~315 words)

Thanks [Member 2]. I'm [Member 3], and I'll cover the most analytically interesting patterns we found — in user activity, time, and content.

Our fourth finding: user posting behaviour follows a classic power-law distribution. Out of 2,346 unique users, the median user posts 2 photos, but the most active user has posted 1,240. Fewer than 5% of users account for the majority of the dataset. This means any model trained on this data without user-level stratification risks learning the behaviour of power users rather than the typical Flickr user.

Our fifth finding was a genuine surprise. Every single photograph in this dataset was taken in 2019 — 87.8% in November and 12.2% in December. This isn't a longitudinal sample of Flickr activity. It's a two-month snapshot. And when we looked at the day of the week, weekends account for 47.2% of all photos taken. The hourly peak is at noon. This is recreational, not professional photography.

Our seventh finding is counter-intuitive. When we broke users into Casual, Moderate, and Active tiers, we expected Active users — the power users — to provide the richest metadata. The opposite is true. Active users write descriptions on only 34.9% of posts, compared to 49.8% for Casual users. The users generating the most content are producing the least context around it. This has direct implications for NLP modelling.

Our tenth finding connects these threads. The top 20 tags are dominated by European place names — but basketball ranks sixth with 1,322 instances, and "basket" ranks eighth with 1,241. Combined with the pan-European geographic spread and the November–December 2019 window, the evidence strongly points to the dataset capturing a specific pan-European basketball tournament. This reframes how we interpret every other finding — this is an event-driven community, not a general Flickr sample.

I'll now hand over to [Member 4], who'll cover our machine learning research questions and reflections on the analysis.

🎤 MEMBER 4 — ML Questions, Reflection, Limitations & Close

Target: 2:30 minutes (~350 words)

Thanks [Member 3]. I'm [Member 4], and I'll walk you through the five ML questions we derived from these findings, and then reflect on the limitations of our analysis.

Our findings led to five ML questions. **Question one** asks whether we can predict missing country values using coordinates and tags — framed as a Random Forest classification problem, with success defined as macro-F1 above 0.75 with balanced recall across the top ten countries. **Question two** asks whether user activity tier — Casual, Moderate, or Active — can be predicted from per-post features, using a multi-class classifier with macro-F1 above 0.70. **Question three** asks whether posting delay can be predicted using a regression model such as a Gradient Boosted Regressor, with success defined as beating the naïve median baseline and achieving R-squared above 0.30. **Question four** asks whether unsupervised clustering on tag content and coordinates can separate event-driven from general photography, with success measured by a silhouette score above 0.3 and interpretable cluster centroids. And **question five** asks whether we can predict whether a post will have a description — targeting ROC-AUC above 0.75 on a user-stratified split to avoid leakage from power users.

Now, reflecting on limitations. First, this dataset is event-specific — a two-month window around a sporting event. Any model we train will carry that selection bias and won't generalise to Flickr platform-wide behaviour. Second, the 57.6% description missingness fundamentally constrains NLP-based modelling — we either impute heavily, or we drop most of the dataset. Third, the power-law user distribution means naïve random splits will leak information between train and test, so user-stratified splitting is essential. Fourth, the multilingual country naming must be normalised through entity resolution before any country-level model is trained.

For improvements, we'd recommend: extending the collection window to capture seasonality, enriching the dataset with sentiment or behavioural signals to enable broader modelling, and building a country-name normalisation layer as a standard preprocessing step.

To conclude, we've presented ten findings, ten insights, and five ML questions — each grounded directly in observed characteristics of the data. Thank you for watching.
